Dynamic Causal Mediation Analysis for Micro-Randomized Trials

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Outline

Motivation: mHealth and Micro-Randomized Trial (MRT)

Solving Challenge 1: Causal Inference with Many Treatment Occasions

Solving Challenge 2: Decomposition into Meaningful Mediated Pathways

A Multiply Robust Estimator

Application: HeartSteps MRT

Mobile Health Interventions

- **mHealth apps:** fitness tracker, mindfulness, medication reminder, drinking diary



(example mHealth apps)

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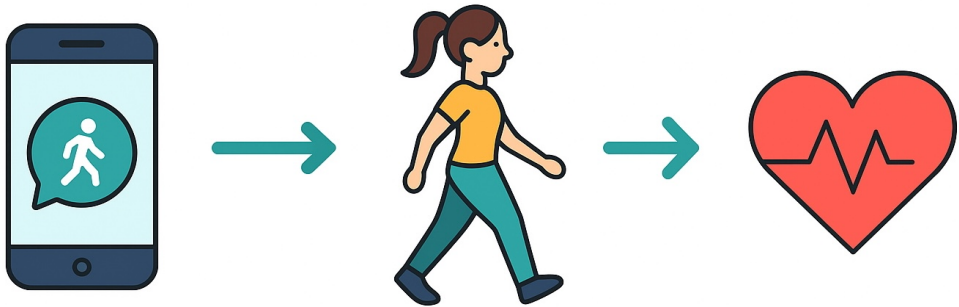
- Underlying premise in mHealth research:

  prompt $\rightarrow$ **short-term behavior change** $\rightarrow$ **long-term health benefits**



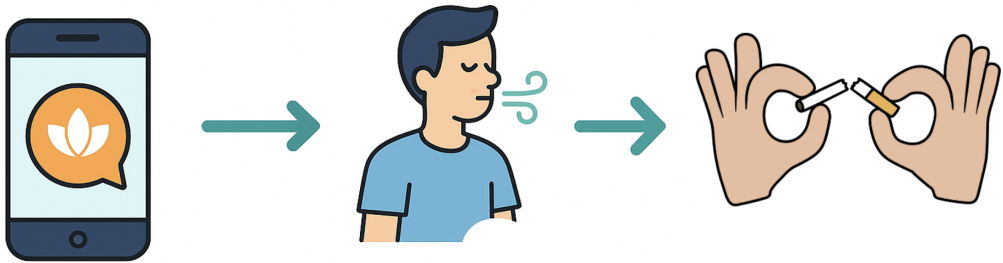
(example mHealth apps)

mHealth Intervention for Physical Activity



activity prompt $\longrightarrow$ physical activity $\longrightarrow$ cardiovascular health

mHealth Intervention for Smoking Cessation



coping prompt $\longrightarrow$ mindfulness exercise $\longrightarrow$ smoking cessation

From Short-term to Long-term Benefits

Underlying premise in mHealth research:

$$\textbf{prompt} \rightarrow \textbf{short-term behavior change} \rightarrow \textbf{long-term health benefits}$$

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Can we verify this premise empirically?

HeartSteps Micro-Randomized Trial

- Micro-randomized trial (MRT) of smartphone prompts to encourage walking
  - $n = 37$ participants, each randomized for $T = 210$ times during the 6-week MRT

(Klasnja et al. 2019)

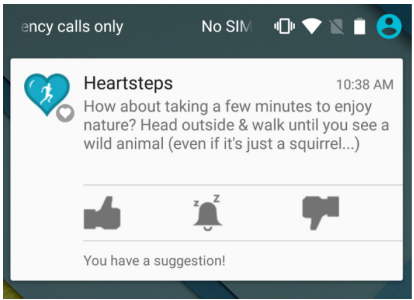
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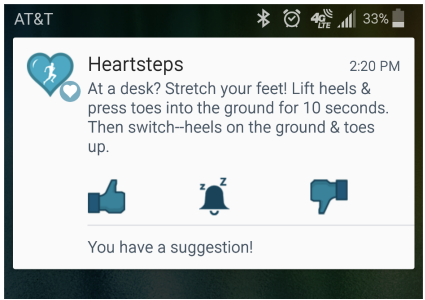
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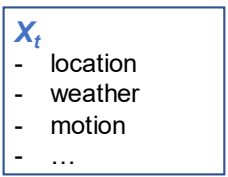
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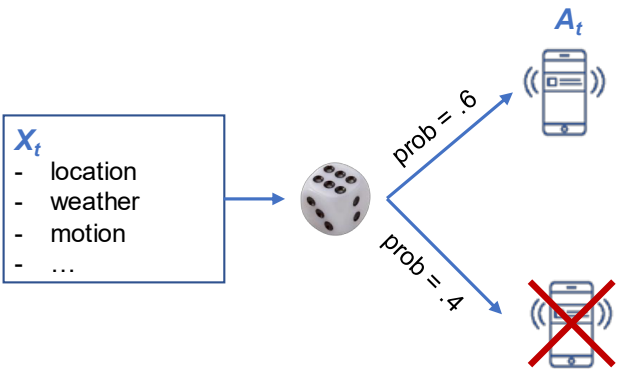
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(Data from an individual at decision point t, with $t = 1, 2, \ldots, 210$)

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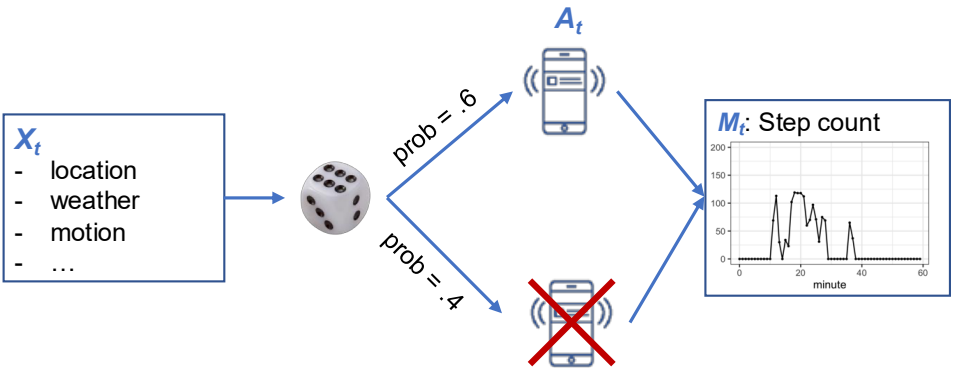
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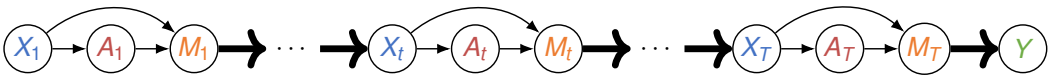
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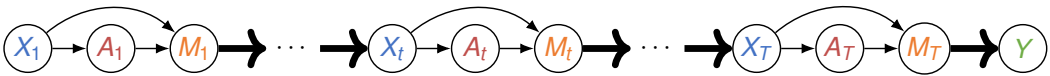
Notation



(Thick arrow denotes arrows from all nodes on the left to all nodes on the right; *no Markovian assumption*.)

	Definition	**Example in HeartSteps**
X_t	Time-varying covariates	Sensor and self-report data
$A_t \in \{0, 1\}$	Time-varying treatment	Activity prompts (or nothing)
M_t	Time-varying mediator	Short-term step count (next 30min)
Y	Distal outcome	Long-term health (end-of-study)

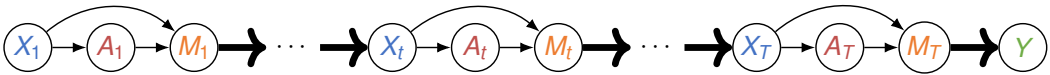
Causal Questions



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- **Short-term effect:** Does A_t influence M_t?
  - Literature: semiparametric models on proximal causal excursion effect (Boruvka et al. 2018; Qian et al. 2021; Shi and Dempsey 2025)

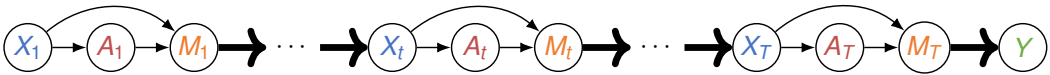
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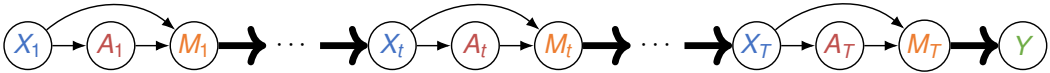
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- **Mediated effect:** Does A_t influence Y *via* M_t? — this talk

Two Challenges

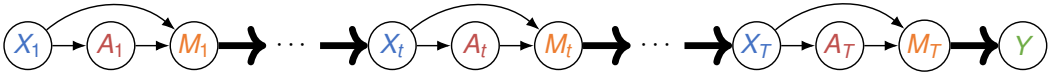


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Challenge 2: Many mediators

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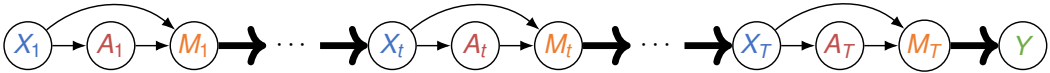
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HeartSteps has 210 decision points. Y may be influenced by *all* $A_1, \ldots, A_{210}$.

How to define and model the treatment effect?

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Challenge 2: Many mediators

For any treatment A_t, its effect on Y may be mediated through *all* $M_t, \ldots, M_{210}$.

How to disentangle the pathways while keeping the estimand interpretable?

Why These Challenges Arise: MRT vs. Classical RCT

	Classical RCT	**Micro-Randomized Trial (MRT)**
Goal	Confirm treatment efficacy	Optimize *when* and *in what context* to intervene
Randomize	Once per person (at baseline)	Hundreds of times per person
Question	"Does Drug A beat placebo?"	"When is a prompt effective? Does the effect change over time?"
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- MRT provides data to build decision rules for sequencing treatments optimally

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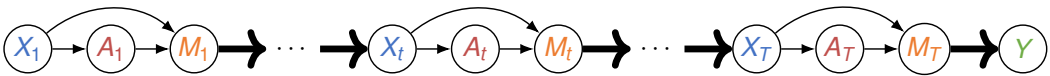
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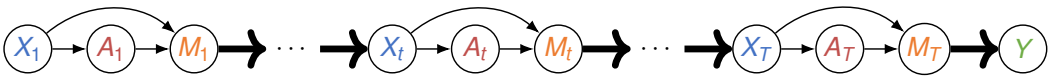
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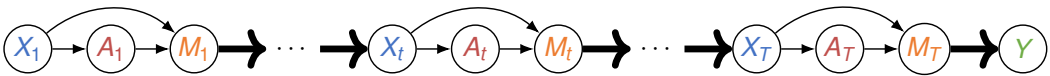
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- Challenge: T is usually comparable to or larger than n
  - $n = 37$, $T = 210$ in HeartSteps

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Potential outcome notation:

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Marginal Contrasts: The “Excursion” Idea

id	A_1	A_2	A_3	A_4	A_5	Y
a	1	0	1	0	1	
b						
c						
d						

id	A_1	A_2	A_3	A_4	A_5	Y
e						
f						
g						
h						

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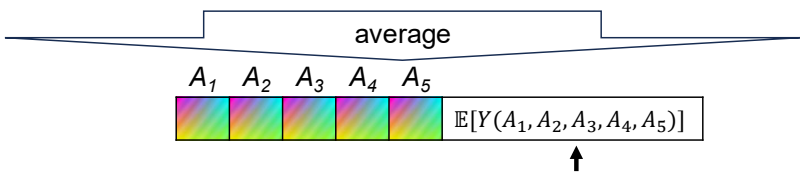
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rainbow color represents MRT policy

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1	0	1	0	1	
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A_1	A_2	A_3	A_4	A_5	Y
0	0	0	1	0	
1	0	0	0	1	
1	0	0	1	1	
1	1	0	1	0	

average

A_1	A_2	A_3	A_4	A_5	
		1			$\mathbb{E}[Y(A_1, A_2, 1, A_4, A_5)]$

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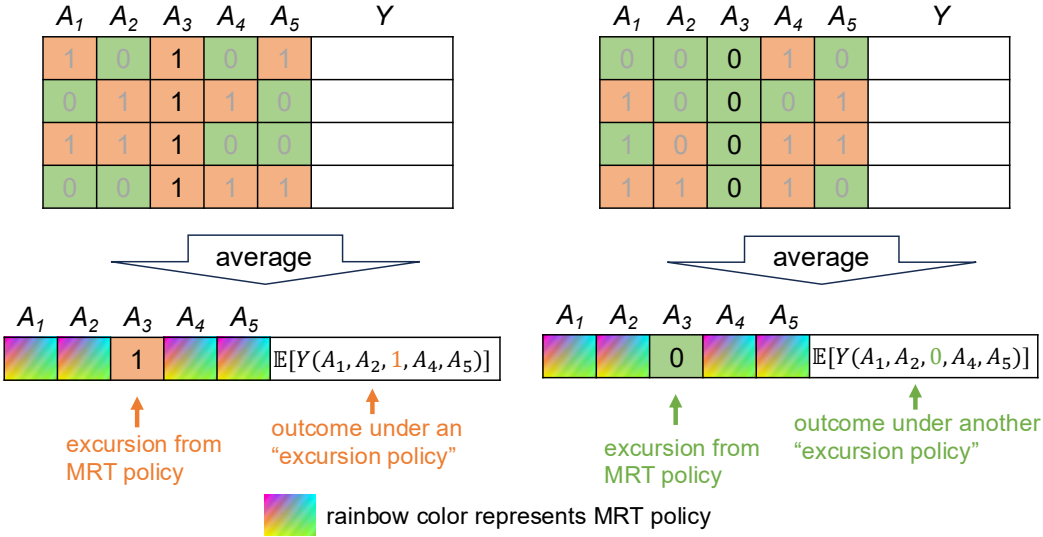
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excursion from MRT policy

outcome under an “excursion policy”

rainbow color represents MRT policy

Marginal Contrasts: The “Excursion” Idea



Total Excursion Effect of A_t on Y

$$\text{TE}_t = \mathbb{E}\big[Y(\overline{A}_{1:t-1}, 1, \overline{A}_{t+1:T}) - Y(\overline{A}_{1:t-1}, 0, \overline{A}_{t+1:T})\big]$$

- Contrast expected potential outcomes between two treatment policies:
  - Follow MRT policy up to $t-1$, then set $A_t = 1$, then follow MRT policy again
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- Interpretation: if we **overrode** the app at time t to **definitely send** a prompt, how would that change the end-of-study outcome compared to **not sending** it at t?

Total Excursion Effect of A_t on Y

$$\begin{aligned}\mathrm{TE}_t &= \mathbb{E}\big[Y(\overline{A}_{1:t-1}, 1, \overline{A}_{t+1:T}) - Y(\overline{A}_{1:t-1}, 0, \overline{A}_{t+1:T})\big] \\ &\equiv \mathbb{E}\big[Y^{A_t=1} - Y^{A_t=0}\big]\end{aligned}$$

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- Interpretation: if we **overrode** the app at time t to **definitely send** a prompt, how would that change the end-of-study outcome compared to **not sending** it at t?
- Simplify notation: $Y^{A_t=1}$ and $Y^{A_t=0}$. Implied: all other A_s are naturally assigned according to the MRT design.

Total Excursion Effect of A_t on Y: $\text{TE}_t = \mathbb{E}\left[Y^{A_t=1} - Y^{A_t=0}\right]$

- Consider a setting where adjacent treatments interact negatively (e.g., back-to-back prompts are less effective due to user burden):
  DGM: $A_t \sim \text{Bernoulli}(p_A)$; $Y = \sum_{t=1}^{T} \beta_t A_t - \sum_{t=1}^{T-1} \alpha_t A_t A_{t+1} + \epsilon$ with $\alpha_t \geq 0$

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- Different MRT with a different $p_A \implies$ different TE.
  - e.g., an MRT that sends prompts 60% of the time has a different TE than one that sends prompts 40% of the time
- Takeaway: Excursion effect is policy-dependent — it answers questions about the specific MRT design, not about a universal treatment effect.

Outline

Motivation: mHealth and Micro-Randomized Trial (MRT)

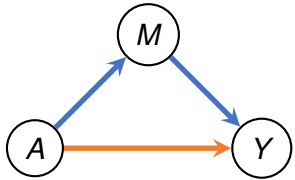
Solving Challenge 1: Causal Inference with Many Treatment Occasions

Solving Challenge 2: Decomposition into Meaningful Mediated Pathways

A Multiply Robust Estimator

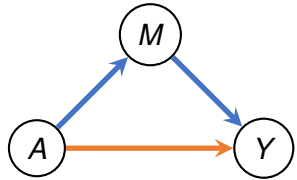
Application: HeartSteps MRT

Review: Causal Mediation Analysis



- Exposure A, mediator M, outcome Y
- A may affect Y directly and indirectly through M

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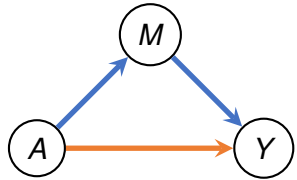
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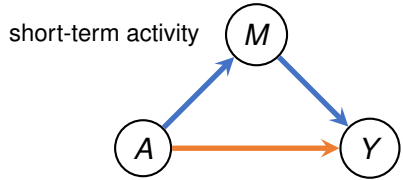
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- e.g., $Y(1, M(0))$: outcome if treated, but mediator behaves as if untreated

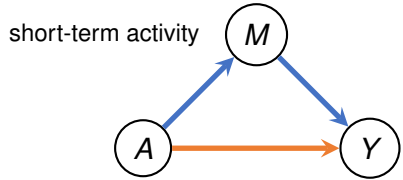
  This is the "cross-world" quantity. We use notation $Y^{A=1 \textit{ but } A=0 \text{ for } M}$

Review: Causal Mediation Analysis



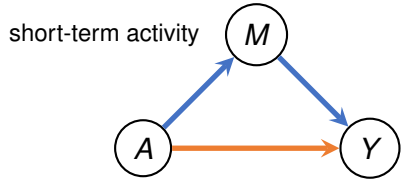
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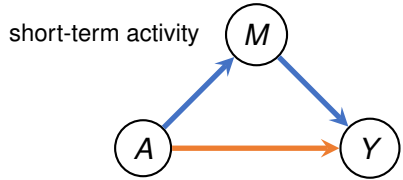
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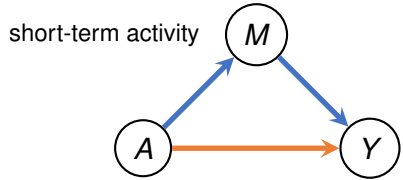
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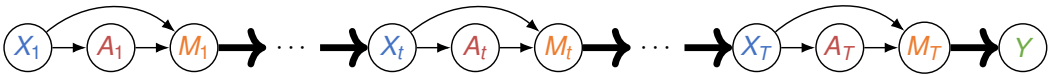
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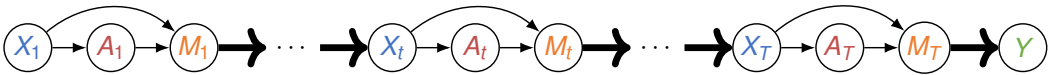
Generalize to Longitudinal Setting



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How to generalize to longitudinal setting with many treatments and many mediators?

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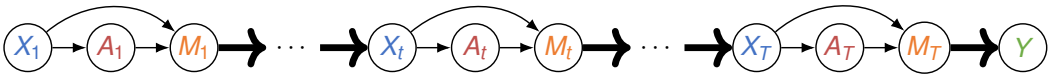
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Challenge 1: Many treatment occasions

Solution: use the excursion effect / marginal contrast idea (already presented)

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How to generalize to longitudinal setting with many treatments and many mediators?

Challenge 1: Many treatment occasions

Solution: use the excursion effect / marginal contrast idea (already presented)

Challenge 2: Many mediators

Solution: for any t, distinguish $A_t \to M_t \to Y$ vs $A_t \to$ other variables $\to Y$ (next)

Excursion Policies and Potential Outcomes

- Excursion policy $D_t^1 := (\overline{A}_{1:t-1}, 1, \overline{A}_{t+1:T})$: Force $A_t = 1$, rest follow MRT policy

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Excursion Policies and Potential Outcomes

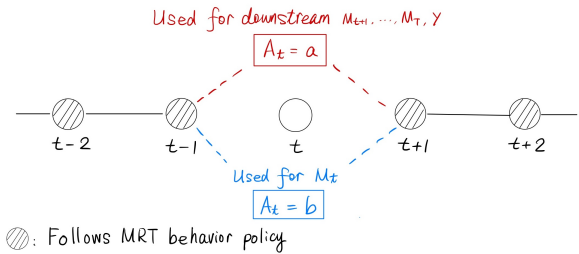
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- To decompose TE into direct and indirect parts, we need a "cross-world" quantity:

  for $a, b \in \{0, 1\}$,

$$\theta_t^{ab} := \mathbb{E}\big[Y\{D_t^a, M_{1:t-1}(D_t^a), M_t(D_t^b), M_{t+1:T}(D_t^a)\}\big]$$

  We use notation: $\mathbb{E}[Y^{A_t=a \text{ but } A_t=b \text{ for } M_t}]$



Indirect and Direct Excursion Effects in MRTs

Total effect: $\text{TE}_t = \theta_t^{11} - \theta_t^{00} = \mathbb{E}\left[Y^{A_t=1}\right] - \mathbb{E}\left[Y^{A_t=0}\right]$

- *HeartSteps: Effect of activity suggestion A_t at decision point t on Y (Week 6 steps)*

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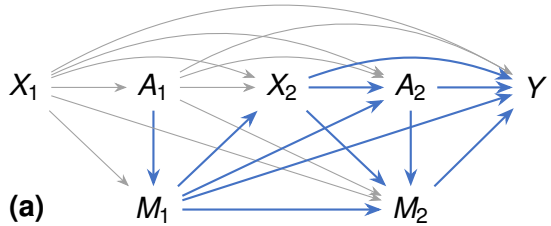
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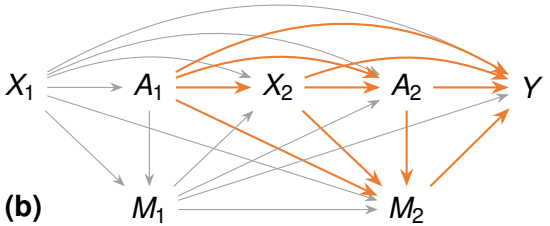
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Example with 2 Time Points

Decomposing the total effect of A_1 on Y into...





- Indirect via M_1
- $\text{IND}_1 = \mathbb{E}\left[Y^{A_1=1}\right] - \mathbb{E}\left[Y^{A_1=1 \textit{ but } A_1=0 \text{ for } M_1}\right]$
- All other paths from A_1 to Y
- $\text{DIR}_1 = \mathbb{E}\left[Y^{A_1=1 \textit{ but } A_1=0 \text{ for } M_1}\right] - \mathbb{E}\left[Y^{A_1=0}\right]$

Why focus on the immediate mediator M_t?

Scientific reason:

Statistical reason:

Causal reason:

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(prompt $\rightarrow$ immediate target $\rightarrow$ long-term goal)

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Causal reason: **improves plausibility of identification assumptions**
(less concern about “intermediate confounding”)

Identification: Causal Assumptions

Under causal assumptions, θ_t^{ab} can be identified from observed data via IPW-type formulas.

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- plausible if (i) we have rich covariate history H_t to adjust for confounding, and (ii) M_t is measured immediately after A_t (e.g., step count in the next 30min)

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- Step 1: estimate $\hat{\theta}_t^{10}$, $\hat{\theta}_t^{00}$, $\hat{\theta}_t^{11}$ for each t (next slides)
- Step 2: regress $\hat{\theta}_t^{10} - \hat{\theta}_t^{00}$ on $f(t)$ to get $\hat{\alpha}$, and regress $\hat{\theta}_t^{11} - \hat{\theta}_t^{10}$ on $f(t)$ to get $\hat{\beta}$

Step 1: Estimate $\theta_t^{11} = \mathbb{E}\left[Y^{A_t=1}\right]$ and $\theta_t^{00} = \mathbb{E}\left[Y^{A_t=0}\right]$

The familiar augmented inverse probability weighting (AIPW) estimating function:

$$\phi_t^{aa}(p_t, \eta_t) := \frac{\mathbb{1}(A_t = a)}{p_t\,(a|H_t)} Y - \frac{\mathbb{1}(A_t = a) - p_t(a|H_t)}{p_t(a|H_t)}\,\eta_t\,(a, H_t)$$

propensity score: $P(A_t = a \mid H_t)$

outcome regression: $\mathbb{E}(Y \mid A_t = a, H_t)$

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outcome regression: $\mathbb{E}(Y \mid A_t = a, H_t)$

Robust in MRTs with known p_t:

$$\underbrace{\mathbb{P}_n\,\phi_t^{aa}(p_t, \hat{\eta}_t)}_{\hat{\theta}_t^{aa}} \approx \theta_t^{aa} \text{ with any } \hat{\eta}_t$$

Step 1: Estimate $\theta_t^{10} = \mathbb{E}\left[Y^{A_t=1 \text{ but } A_t=0 \text{ for } M_t} \right]$

Estimating function:

q_t: propensity score given M_t

μ_t: outcome regression given M_t

$$\phi_t^{ab}(p_t, q_t, \mu_t, \nu_t) := \frac{\mathbb{1}(A_t = a)\, q_t\,(b|H_t, M_t)}{p_t(b|H_t)\, q_t(a|H_t, M_t)} \Big\{ Y - \mu_t\,(a, H_t, M_t) \Big\} + \frac{\mathbb{1}(A_t = b)}{p_t\,(b|H_t)} \Big\{ \mu_t(a, H_t, M_t) - \nu_t\,(a, H_t) \Big\} + \nu_t(a, H_t).$$

ν_t: cross-world outcome regression

p_t: propensity score

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ν_t: cross-world outcome regression

p_t: propensity score

Doubly robust in MRTs with known p_t:

$$\underbrace{\mathbb{P}_n\,\phi_t^{10}(\hat{p}_t, \hat{q}_t, \hat{\mu}_t, \hat{\nu}_t)}_{\hat{\theta}_t^{10}} \approx \theta_t^{10} \text{ if either } \hat{q}_t \text{ or } \hat{\mu}_t \text{ is correct}$$

Robustness of the Estimator $(\hat{\alpha}, \hat{\beta})$

MRT setting (known p_t): Doubly Robust

$(\hat{\alpha}, \hat{\beta})$ is consistent and asymptotically normal if either q_t or μ_t is correctly specified.

- Only need to get *one* of two modeling tasks right

p_t (propensity), q_t (propensity given M_t), η_t (outcome reg.), μ_t (outcome reg. given M_t), ν_t (cross-world reg.)

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Observational study (unknown p_t): Multiply Robust

$(\hat{\alpha}, \hat{\beta})$ is consistent and asymptotically normal if any one of these holds:

(i) p_t and q_t correct

(ii) p_t and μ_t correct

(iii) η_t, ν_t, and q_t correct

(iv) η_t, ν_t, and μ_t correct

- Four chances to get it right — only need one combination

p_t (propensity), q_t (propensity given M_t), η_t (outcome reg.), μ_t (outcome reg. given M_t), ν_t (cross-world reg.)

Outline

Motivation: mHealth and Micro-Randomized Trial (MRT)

Solving Challenge 1: Causal Inference with Many Treatment Occasions

Solving Challenge 2: Decomposition into Meaningful Mediated Pathways

A Multiply Robust Estimator

Application: HeartSteps MRT

Application: HeartSteps Micro-Randomized Trial

- $n = 37$ participants, $T = 175$ decision points in Weeks 1–5 (Week 6 reserved for Y)
- A_t: randomized activity suggestion (5x/day, rand. prob. 0.6)
- M_t: 30-min step count following A_t
- Y: average daily steps in Week 6

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- DIR_t: long-term effect via other pathways (e.g., habit formation)

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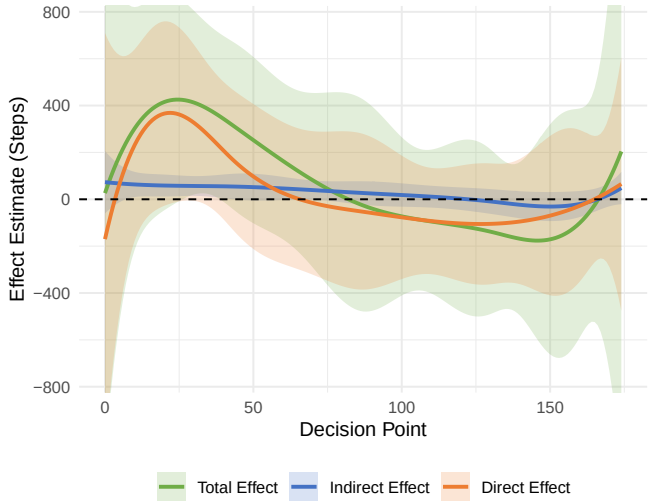
Recall: Direct and Indirect Effects of A_t on Y through M_t

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Model Fitting:

- IND_t and DIR_t modeled using B-spline basis over t
- GAMs used for nuisance functions

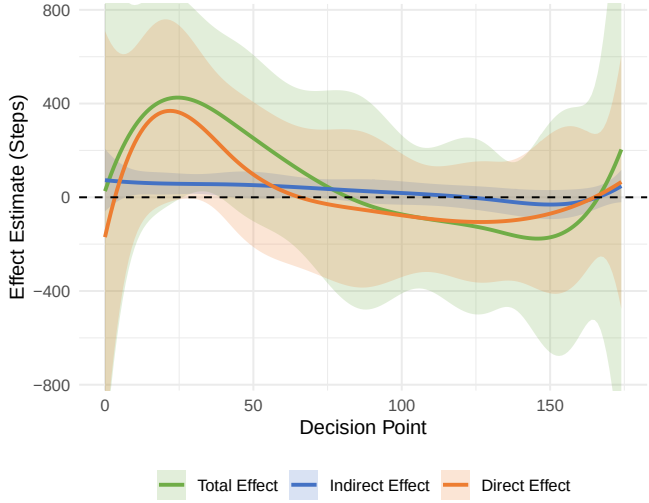
HeartSteps: Effect Estimates Over Time



HeartSteps: Effect Estimates Over Time

Findings:

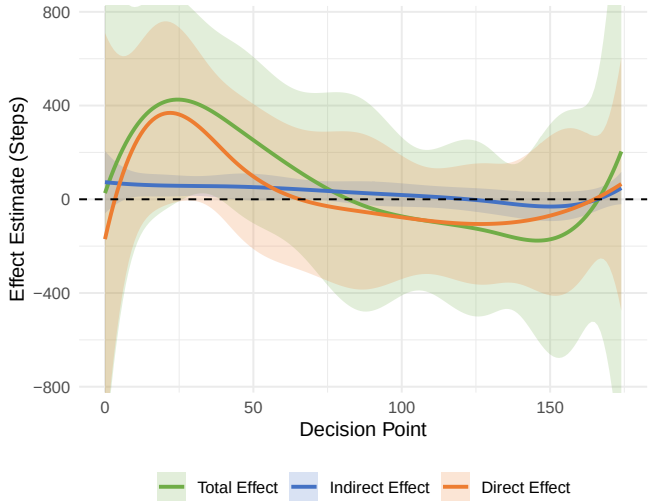
- Early direct effects large but decline rapidly



HeartSteps: Effect Estimates Over Time

Findings:

- Early direct effects large but decline rapidly
- Indirect effects more stable and persistent



HeartSteps: Scientific Takeaway

Two mechanisms:

- Effect of early prompts on Y: through pathways other than immediate walking
- Effect of later prompts on Y: through increasing immediate walking

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1. What constitutes "pathways other than immediate walking"? (motivation? positive habit formation?)
2. Why pathways other than immediate walking no longer effective later in the study? (burden/habituation? habit already formed?)
3. Sensitivity to the cross-world ignorability assumption?

Summary

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Summary

Motivation: Quantify how short-term behavioral responses mediate long-term effects of mHealth interventions

Challenges: Many treatment occasions, many mediators

Contributions:

- Define direct and indirect excursion effects for time-varying mediation
- Multiply robust estimator

Findings from HeartSteps MRT:

- Early direct effects large but decline rapidly
- Indirect effects (through increasing immediate steps) more stable and persistent

Backup: Identification Results

Under assumptions (i)–(v):

- For $a \in \{0, 1\}$ (same-world):

$$\theta_t^{aa} = \mathbb{E}\Big[\frac{\mathbb{1}(A_t = a)}{P(A_t = a \mid H_t)} Y\Big]$$

- For $a, b \in \{0, 1\}$ and $a \neq b$ (cross-world):

$$\theta_t^{ab} = \mathbb{E}\Big[\frac{\mathbb{1}(A_t = a)}{P(A_t = b \mid H_t)} \cdot \frac{P(A_t = b \mid H_t, M_t)}{P(A_t = a \mid H_t, M_t)} Y\Big]$$

Details: Nuisance Functions

Notation	Definition	Description
$p_t^\star(a \mid H_t)$	$P(A_t = a \mid H_t)$	Propensity score
$q_t^\star(a \mid H_t, M_t)$	$P(A_t = a \mid H_t, M_t)$	Propensity score given M_t
$\eta_t^\star(a, H_t)$	$\mathbb{E}(Y \mid A_t = a, H_t)$	Outcome regression
$\mu_t^\star(a, H_t, M_t)$	$\mathbb{E}(Y \mid A_t = a, H_t, M_t)$	Outcome regression given M_t
$\nu_t^\star(a, H_t)$	$\mathbb{E}\{\mu_t^\star(a, H_t, M_t) \mid A_t = 1-a, H_t\}$	Cross-world outcome regression

Backup: Rate Multiple Robustness (nonparametric nuisance fit)

Theorem: The estimator $(\hat{\alpha}, \hat{\beta})$ is CAN if all nuisance parameter estimates converge to the truth at fast enough rates.

Specifically, for each $t = 1, \ldots, T$, we require

$$\|\hat{p}_t - p_t^\star\| \cdot \left(\|\hat{\eta}_t - \eta_t^\star\| + \|\hat{\nu}_t - \nu_t^\star\|\right) = o_P(n^{-\frac{1}{2}})$$

$$\text{and} \qquad \|\hat{q}_t - q_t^\star\| \cdot \|\hat{\mu}_t - \mu_t^\star\| = o_P(n^{-\frac{1}{2}})$$

Nuisance Function Definition

- p_t: $P(A_t \mid H_t)$
- q_t: $P(A_t \mid H_t, M_t)$
- η_t: $\mathbb{E}(Y \mid A_t, H_t)$
- μ_t: $\mathbb{E}(Y \mid A_t, H_t, M_t)$
- ν_t: $\mathbb{E}\{\mu_t \mid A_t = 1 - a, H_t\}$

A sufficient condition: each converges faster than $n^{-1/4}$-rate.

Corollary: For MRT (known p_t), it suffices to have $\|\hat{q}_t - q_t^\star\| \cdot \|\hat{\mu}_t - \mu_t^\star\| = o_P(n^{-\frac{1}{2}})$.

Backup: Efficient Influence Functions

- The efficient influence function (EIF) for θ_t^{aa} is $\phi_t^{aa} - \theta_t^{aa}$.
- The EIF for θ_t^{ab} is $\phi_t^{ab} - \theta_t^{ab}$.
- This means that the estimators

$$\hat{\theta}_t^{11} \stackrel{\text{def}}{=\!=} \mathbb{P}_n \, \phi_t^{11}(\hat{p}_t, \hat{\eta}_t)$$

$$\hat{\theta}_t^{10} \stackrel{\text{def}}{=\!=} \mathbb{P}_n \, \phi_t^{10}(\hat{p}_t, \hat{q}_t, \hat{\mu}_t, \hat{\nu}_t)$$

$$\hat{\theta}_t^{00} \stackrel{\text{def}}{=\!=} \mathbb{P}_n \, \phi_t^{00}(\hat{p}_t, \hat{\eta}_t)$$

  have the smallest asymptotic variance among all regular and asymptotically linear estimators when all nuisance parameters are correctly modeled.

Backup: Estimation Procedure (Formal)

Estimand: (α, β). Assume parametric models for the (in)direct effects:

$$\text{DIR}_t \overset{\text{def}}{=\!=} \theta_t^{10} - \theta_t^{00} \overset{\text{model}}{=\!=} f(t)^\top \alpha$$

$$\text{IND}_t \overset{\text{def}}{=\!=} \theta_t^{11} - \theta_t^{10} \overset{\text{model}}{=\!=} f(t)^\top \beta$$

Estimation: (formal)

- First, fit the five nuisance functions $\{p_t, q_t, \mu_t, \nu_t, \eta_t\}$ using parametric or nonparametric (machine learning) methods
- Then, solve $(\hat{\alpha}, \hat{\beta})$ from

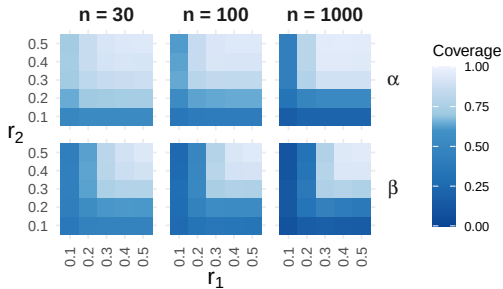
$$\mathbb{P}_n \sum_{t=1}^{T} \begin{bmatrix} \left\{\phi_t^{10}(\hat{p}_t, \hat{q}_t, \hat{\mu}_t, \hat{\nu}_t) - \phi_t^{00}(\hat{p}_t, \hat{\eta}_t) - f(t)^\top \alpha\right\} f(t) \\ \left\{\phi_t^{11}(\hat{p}_t, \hat{\eta}_t) - \phi_t^{10}(\hat{p}_t, \hat{q}_t, \hat{\mu}_t, \hat{\nu}_t) - f(t)^\top \beta\right\} f(t) \end{bmatrix} = 0$$

Backup: Simulation 1 (Verifying Rate Multiple Robustness)

Design:

- $T = 5$; true nuisance functions known in closed form
- Perturb each truth to control its rate, e.g., $\hat{p}_t = U_n p_t^\star$, $U_n \sim \text{Unif}[1 - n^{-r}, 1]$
- r_1: rate of $\hat{p}_t, \hat{\eta}_t, \hat{\nu}_t$
  r_2: rate of $\hat{q}_t, \hat{\mu}_t$

Theory: CAN if $r_1 > 1/4$ and $r_2 > 1/4$



Coverage of 95% CI for $\hat{\alpha}$ (direct) and $\hat{\beta}$ (indirect)

- Near 95% coverage when $r_1, r_2 \geq 0.3$
- Collapses as n grows if either rate ≤ 0.2

Backup: Simulation 2 (Realistic MRT-Like Setting)

Design: $T = 30$, serial correlation, time-varying availability, nonlinear effects

- p_t known (as in an MRT); $\eta_t = \nu_t = 0$ (severely misspecified) in all scenarios
- Vary the working models (GAMs) for q_t and μ_t:

Scenario	Model for q_t	Model for μ_t	Covered by theory?
1	correct	moderately misspecified	yes
2	severely misspecified	moderately misspecified	no
3	correct	severely misspecified	yes
4	severely misspecified	severely misspecified	no

Correct: $s(t)+A_{t-1}+s(M_{t-1})+s(X_t)+s(M_t)$ Moderate: $s(t)+s(X_t)+s(M_t)$ Severe: $s(t)$

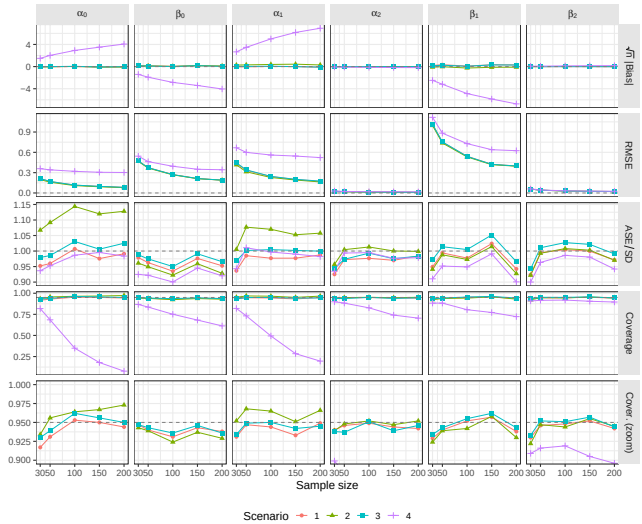
Recall: with known p_t, the estimator is doubly robust (need either q_t or μ_t correct)

Backup: Simulation 2 Results

- Scenarios 1 & 3 (covered by theory): negligible bias, ASE/SD ≈ 1, nominal coverage
- Scenario 2: slight overcoverage
- Scenario 4 (both q_t and μ_t severely wrong): $\sqrt{n}\,|\text{Bias}|$ grows, coverage collapses

α_0, β_0: $f(t) = 1$

$(\alpha_1, \alpha_2), (\beta_1, \beta_2)$: $f(t) = (1, t)^\top$



Backup: Simulation 2 Generative Model

$T = 30$; for $t = 1, \ldots, T$, generate sequentially (I_t: availability indicator):

$$X_t \sim N\big(0.3X_{t-1} + 0.2A_{t-1} + 0.2M_{t-1}, 1\big)$$

$$I_t \sim \text{Bern}\big\{\text{expit}(1.5 - 0.3A_{t-1} - 0.3M_{t-1} + 0.3X_t)\big\}$$

$$A_t \sim \text{Bern}\big[I_t \,\text{expit}\{0.2A_{t-1} + 0.2h(t, M_{t-1}) + 0.3h(t, X_t)\}\big]$$

$$M_t \sim N\big\{0.4A_{t-1} + 0.4h(t, M_{t-1}) + 0.3h(t, X_t) + 0.6A_t, 1\big\}$$

$$Y \sim N\Big[\textstyle\sum_{t=1}^{T}\big\{0.3h(t, X_t) + 0.4h(t, M_t) + 0.2A_t + 0.1A_t h(t, M_t)\big\}, 1\Big]$$

with $h(t, z) := \tanh\{3(2t - T)/T\} + \sin(z)$, nonlinear in both arguments

- $q_t^\star$ depends only on $(t, A_{t-1}, M_{t-1}, X_t, M_t)$: well approximated by a GAM
- $\eta_t^\star, \mu_t^\star, \nu_t^\star$ depend on the full history H_t: correct specification infeasible

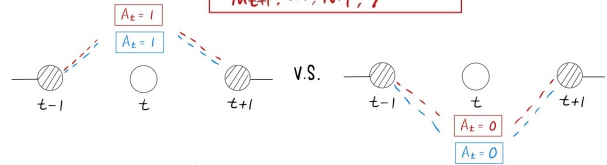
Backup: (In)Direct Excursion Effects

Used for downstream $M_{t+1}, \cdots, M_T, Y$

Used for M_t

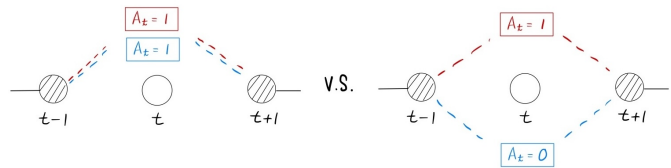
Total Excursion Effect:

$\text{TE}_t := \theta_t^{11} - \theta_t^{00}$



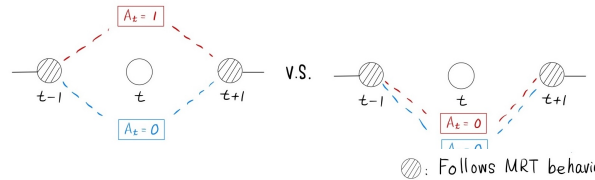
Natural Indirect Excursion Effect:

$\text{IND}_t := \theta_t^{11} - \theta_t^{10}$



Natural Direct Excursion Effect:

$\text{DIR}_t := \theta_t^{10} - \theta_t^{00}$



In HeartSteps: TE_t = effect via immediate walking (IND_t) + effect via other pathways (DIR_t)

Backup: Multiple Robustness (parametric nuisance fit)

Theorem: The estimator $(\hat{\alpha}, \hat{\beta})$ is CAN if for each $t = 1, \ldots, T$, any of the following four combinations is correctly modeled parametrically:

(i) p_t and q_t; or

(ii) p_t and μ_t; or

(iii) η_t, ν_t and q_t; or

(iv) η_t, ν_t and μ_t

Nuisance Function Definition

p_t: $P(A_t \mid H_t)$

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ν_t: $\mathbb{E}\{\mu_t \mid A_t = 1 - a, H_t\}$

Corollary: For MRT (known p_t), it suffices to have either q_t or μ_t correct.

Reference I

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Reference II

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